

✓ Solution

Ⓐ To evaluate $-x$ when $x = 8$, substitute 8 for x .

	$-x$
Substitute 8 for x .	$-(8)$
Simplify.	-8

Ⓑ To evaluate $-x$ when $x = -8$, substitute -8 for x .

	$-x$
Substitute -8 for x .	$-(-8)$
Simplify.	8

> **TRY IT :: 3.9** Evaluate $-n$:

- Ⓐ when $n = 4$ Ⓑ when $n = -4$

> **TRY IT :: 3.10** Evaluate: $-m$:

- Ⓐ when $m = 11$ Ⓑ when $m = -11$

Simplify Expressions with Absolute Value

We saw that numbers such as 5 and -5 are opposites because they are the same distance from 0 on the number line. They are both five units from 0. The distance between 0 and any number on the number line is called the **absolute value** of that number. Because distance is never negative, the absolute value of any number is never negative.

The symbol for absolute value is two vertical lines on either side of a number. So the absolute value of 5 is written as $|5|$, and the absolute value of -5 is written as $|-5|$ as shown in Figure 3.16.

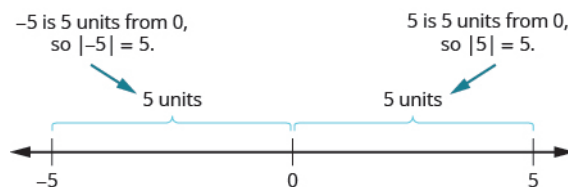


Figure 3.16

Absolute Value

The absolute value of a number is its distance from 0 on the number line.

The absolute value of a number n is written as $|n|$.

$$|n| \geq 0 \text{ for all numbers}$$

EXAMPLE 3.6

Simplify:

- Ⓐ $|3|$ Ⓑ $|-44|$ Ⓒ $|0|$

✓ **Solution**

Ⓐ

	$ 3 $
3 is 3 units from zero.	3

Ⓑ

	$ -44 $
-44 is 44 units from zero.	44

Ⓒ

	$ 0 $
0 is already at zero.	0

> **TRY IT :: 3.11** Simplify:

- Ⓐ $|12|$ Ⓑ $-|-28|$

> **TRY IT :: 3.12** Simplify:

- Ⓐ $|9|$ Ⓑ $(b) - |37|$

We treat absolute value bars just like we treat parentheses in the order of operations. We simplify the expression inside first.

EXAMPLE 3.7

Evaluate:

- Ⓐ $|x|$ when $x = -35$ Ⓑ $|-y|$ when $y = -20$ Ⓒ $-|u|$ when $u = 12$ Ⓓ $-|p|$ when $p = -14$

✓ **Solution**

Ⓐ To find $|x|$ when $x = -35$:

	$ x $
Substitute -35 for x .	$ -35 $
Take the absolute value.	35

ⓑ To find $|-y|$ when $y = -20$:

	$ -y $
Substitute -20 for y .	$ -(-20) $
Simplify.	$ 20 $
Take the absolute value.	20

ⓒ To find $-|u|$ when $u = 12$:

	$- u $
Substitute 12 for u .	$- 12 $
Take the absolute value.	-12

ⓓ To find $-|p|$ when $p = -14$:

	$- p $
Substitute -14 for p .	$- -14 $
Take the absolute value.	-14

Notice that the result is negative only when there is a negative sign outside the absolute value symbol.

> TRY IT :: 3.13

Evaluate:

- ⓐ $|x|$ when $x = -17$ ⓑ $|-y|$ when $y = -39$ ⓒ $-|m|$ when $m = 22$
 ⓓ $-|p|$ when $p = -11$

> TRY IT :: 3.14

- ⓐ $|y|$ when $y = -23$ ⓑ $|-y|$ when $y = -21$ ⓒ $-|n|$ when $n = 37$
 ⓓ $-|q|$ when $q = -49$

EXAMPLE 3.8

Fill in $<$, $>$, or $=$ for each of the following:

- ⓐ $|-5|$ ___ $-|-5|$ ⓑ 8 ___ $-|-8|$ ⓒ -9 ___ $-|-9|$ ⓓ $-|-7|$ ___ -7

✓ Solution

To compare two expressions, simplify each one first. Then compare.

Ⓐ

$$|-5| \underline{\hspace{1cm}} - |-5|$$

$$\text{Simplify. } 5 \underline{\hspace{1cm}} -5$$

$$\text{Order. } 5 > -5$$

Ⓑ

$$8 \underline{\hspace{1cm}} - |-8|$$

$$\text{Simplify. } 8 \underline{\hspace{1cm}} - 8$$

$$\text{Order. } 8 > -8$$

Ⓒ

$$-9 \underline{\hspace{1cm}} - |-9|$$

$$\text{Simplify. } -9 \underline{\hspace{1cm}} - 9$$

$$\text{Order. } -9 = -9$$

Ⓓ

$$- |-7| \underline{\hspace{1cm}} -7$$

$$\text{Simplify. } -7 \underline{\hspace{1cm}} - 7$$

$$\text{Order. } -7 = -7$$

**TRY IT :: 3.15**Fill in $<$, $>$, or $=$ for each of the following:

Ⓐ $|-9| \underline{\hspace{1cm}} - |-9|$

Ⓑ $2 \underline{\hspace{1cm}} - |-2|$

Ⓒ $-8 \underline{\hspace{1cm}} |-8|$

Ⓓ $- |-5| \underline{\hspace{1cm}} -5$

**TRY IT :: 3.16**Fill in $<$, $>$, or $=$ for each of the following:

Ⓐ $7 \underline{\hspace{1cm}} - |-7|$

Ⓑ $- |-11| \underline{\hspace{1cm}} -11$

Ⓒ $|-4| \underline{\hspace{1cm}} - |-4|$

Ⓓ $-1 \underline{\hspace{1cm}} |-1|$

Absolute value bars act like grouping symbols. First simplify inside the absolute value bars as much as possible. Then take the absolute value of the resulting number, and continue with any operations outside the absolute value symbols.

EXAMPLE 3.9

Simplify:

Ⓐ $|9-3|$ Ⓑ $4|-2|$

✓ **Solution**

For each expression, follow the order of operations. Begin inside the absolute value symbols just as with parentheses.

Ⓐ

	$ 9-3 $
Simplify inside the absolute value sign.	$ 6 $
Take the absolute value.	6

Ⓑ

	$4 -2 $
Take the absolute value.	$4 \cdot 2$
Multiply.	8

> **TRY IT :: 3.17** Simplify:

Ⓐ $|12 - 9|$ Ⓑ $3|-6|$

> **TRY IT :: 3.18** Simplify:

Ⓐ $|27 - 16|$ Ⓑ $9|-7|$

EXAMPLE 3.10

Simplify: $|8 + 7| - |5 + 6|$.

✓ **Solution**

For each expression, follow the order of operations. Begin inside the absolute value symbols just as with parentheses.

	$ 8+7 - 5+6 $
Simplify inside each absolute value sign.	$ 15 - 11 $
Subtract.	4

> **TRY IT :: 3.19** Simplify: $|1 + 8| - |2 + 5|$

> **TRY IT :: 3.20** Simplify: $|9-5| - |7 - 6|$

EXAMPLE 3.11

Simplify: $24 - |19 - 3(6 - 2)|$.

✓ Solution

We use the order of operations. Remember to simplify grouping symbols first, so parentheses inside absolute value symbols would be first.

	$24 - 19 - 3(6 - 2) $
Simplify in the parentheses first.	$24 - 19 - 3(4) $
Multiply $3(4)$.	$24 - 19 - 12 $
Subtract inside the absolute value sign.	$24 - 7 $
Take the absolute value.	$24 - 7$
Subtract.	17

> **TRY IT :: 3.21** Simplify: $19 - |11 - 4(3 - 1)|$

> **TRY IT :: 3.22** Simplify: $9 - |8 - 4(7 - 5)|$

Translate Word Phrases into Expressions with Integers

Now we can translate word phrases into expressions with integers. Look for words that indicate a negative sign. For example, the word *negative* in “negative twenty” indicates -20 . So does the word *opposite* in “the opposite of 20.”

EXAMPLE 3.12

Translate each phrase into an expression with integers:

- (a) the opposite of positive fourteen (b) the opposite of -11 (c) negative sixteen
 (d) two minus negative seven

✓ Solution

- (a) the opposite of fourteen (b) the opposite of -11 (c) negative sixteen (d) two minus negative seven
 -14 $-(-11)$ -16 $2 - (-7)$

> **TRY IT :: 3.23**

Translate each phrase into an expression with integers:

- (a) the opposite of positive nine (b) the opposite of -15 (c) negative twenty
 (d) eleven minus negative four

> **TRY IT :: 3.24**

Translate each phrase into an expression with integers:

- (a) the opposite of negative nineteen (b) the opposite of twenty-two (c) negative nine
 (d) negative eight minus negative five

As we saw at the start of this section, negative numbers are needed to describe many real-world situations. We'll look at some more applications of negative numbers in the next example.

EXAMPLE 3.13

Translate into an expression with integers:

- Ⓐ The temperature is 12 degrees Fahrenheit below zero. Ⓑ The football team had a gain of 3 yards.
- Ⓒ The elevation of the Dead Sea is 1,302 feet below sea level.
- Ⓓ A checking account is overdrawn by \$40.

✓ **Solution**

Look for key phrases in each sentence. Then look for words that indicate negative signs. Don't forget to include units of measurement described in the sentence.

Ⓐ The temperature is 12 degrees Fahrenheit below zero.

Below zero tells us that 12 is a negative number. -12°F

Ⓑ The football team had a gain of 3 yards.

A *gain* tells us that 3 is a positive number. 3 yards

Ⓒ The elevation of the Dead Sea is 1,302 feet below sea level.

Below sea level tells us that 1,302 is a negative number. $-1,302$ feet

Ⓓ A checking account is overdrawn by \$40.

Overdrawn tells us that 40 is a negative number. $-\$40$

> **TRY IT :: 3.25** Translate into an expression with integers:
The football team had a gain of 5 yards.

> **TRY IT :: 3.26** Translate into an expression with integers:
The scuba diver was 30 feet below the surface of the water.

▶ **MEDIA :: ACCESS ADDITIONAL ONLINE RESOURCES**

- **Introduction to Integers** (<http://openstaxcollege.org/l/24introinteger>)
- **Simplifying the Opposites of Negative Integers** (<http://openstaxcollege.org/l/24neginteger>)
- **Comparing Absolute Value of Integers** (<http://openstaxcollege.org/l/24abvalue>)
- **Comparing Integers Using Inequalities** (<http://openstaxcollege.org/l/24usinginequal>)



3.1 EXERCISES

Practice Makes Perfect

Locate Positive and Negative Numbers on the Number Line

In the following exercises, locate and label the given points on a number line.

1.

- (a) 2 (b) -2
(c) -5

2.

- (a) 5 (b) -5
(c) -2

3.

- (a) -8 (b) 8
(c) -6

4.

- (a) -7 (b) 7
(c) -1

Order Positive and Negative Numbers on the Number Line

In the following exercises, order each of the following pairs of numbers, using $<$ or $>$.

5.

- (a) 9__4 (b) -3__6
(c) -8__-2 (d) 1__-10

6.

- (a) 6__2; (b) -7__4;
(c) -9__-1; (d) 9__-3

7.

- (a) -5__1; (b) -4__-9;
(c) 6__10; (d) 3__-8

8.

- (a) -7__3;
(b) -10__-5;
(c) 2__-6; (d) 8__9

Find Opposites

In the following exercises, find the opposite of each number.

9.

- (a) 2 (b) -6

10.

- (a) 9 (b) -4

11.

- (a) -8 (b) 1

12.

- (a) -2 (b) 6

In the following exercises, simplify.

13. $-(-4)$

14. $-(-8)$

15. $-(-15)$

16. $-(-11)$

In the following exercises, evaluate.

17. $-m$ when

- (a) $m = 3$ (b) $m = -3$

18. $-p$ when

- (a) $p = 6$ (b) $p = -6$

19. $-c$ when

- (a) $c = 12$ (b) $c = -12$

20. $-d$ when

- (a) $d = 21$
(b) $d = -21$

Simplify Expressions with Absolute Value*In the following exercises, simplify each absolute value expression.***21.**

- Ⓐ $|7|$ Ⓑ $|-25|$
 Ⓒ $|0|$

22.

- Ⓐ $|5|$ Ⓑ $|20|$
 Ⓒ $|-19|$

23.

- Ⓐ $|-32|$ Ⓑ $|-18|$
 Ⓒ $|16|$

24.

- Ⓐ $|-41|$ Ⓑ $|-40|$
 Ⓒ $|22|$

*In the following exercises, evaluate each absolute value expression.***25.**

- Ⓐ $|x|$ when $x = -28$
 Ⓑ $|-u|$ when $u = -15$

26.

- Ⓐ $|y|$ when $y = -37$
 Ⓑ $|-z|$ when $z = -24$

27.

- Ⓐ $-|p|$ when $p = 19$
 Ⓑ $-|q|$ when $q = -33$

28.

- Ⓐ $-|a|$ when $a = 60$
 Ⓑ $-|b|$ when $b = -12$

*In the following exercises, fill in $<$, $>$, or $=$ to compare each expression.***29.**

- Ⓐ $-6 \underline{\hspace{1cm}} |-6|$
 Ⓑ $-|-3| \underline{\hspace{1cm}} -3$

30.

- Ⓐ $-8 \underline{\hspace{1cm}} |-8|$
 Ⓑ $-|-2| \underline{\hspace{1cm}} -2$

31.

- Ⓐ $|-3| \underline{\hspace{1cm}} -|-3|$
 Ⓑ $4 \underline{\hspace{1cm}} -|-4|$

32.

- Ⓐ $|-5| \underline{\hspace{1cm}} -|-5|$
 Ⓑ $9 \underline{\hspace{1cm}} -|-9|$

In the following exercises, simplify each expression.

33. $|8 - 4|$

34. $|9 - 6|$

35. $8|-7|$

36. $5|-5|$

37. $|15 - 7| - |14 - 6|$

38. $|17 - 8| - |13 - 4|$

39. $18 - |2(8 - 3)|$

40. $15 - |3(8 - 5)|$

41. $8(14 - 2|-2|)$

42. $6(13 - 4|-2|)$

Translate Word Phrases into Expressions with Integers*Translate each phrase into an expression with integers. Do not simplify.***43.**

- Ⓐ the opposite of 8
 Ⓑ the opposite of -6
 Ⓒ negative three
 Ⓓ 4 minus negative 3

44.

- Ⓐ the opposite of 11
 Ⓑ the opposite of -4
 Ⓒ negative nine
 Ⓓ 8 minus negative 2

45.

- Ⓐ the opposite of 20
 Ⓑ the opposite of -5
 Ⓒ negative twelve
 Ⓓ 18 minus negative 7

46.
 Ⓐ the opposite of 15
 Ⓑ the opposite of -9
 Ⓒ negative sixty
 Ⓓ 12 minus 5
47. a temperature of 6 degrees below zero
48. a temperature of 14 degrees below zero
49. an elevation of 40 feet below sea level
50. an elevation of 65 feet below sea level
51. a football play loss of 12 yards
52. a football play gain of 4 yards
53. a stock gain of \$3
54. a stock loss of \$5
55. a golf score one above par
56. a golf score of 3 below par

Everyday Math

57. Elevation The highest elevation in the United States is Mount McKinley, Alaska, at 20,320 feet above sea level. The lowest elevation is Death Valley, California, at 282 feet below sea level. Use integers to write the elevation of:

- Ⓐ Mount McKinley Ⓑ Death Valley

59. State budgets In June, 2011, the state of Pennsylvania estimated it would have a budget surplus of \$540 million. That same month, Texas estimated it would have a budget deficit of \$27 billion. Use integers to write the budget:

- Ⓐ surplus Ⓑ deficit

58. Extreme temperatures The highest recorded temperature on Earth is 58° Celsius, recorded in the Sahara Desert in 1922. The lowest recorded temperature is 90° below 0° Celsius, recorded in Antarctica in 1983. Use integers to write the:

- Ⓐ highest recorded temperature
 Ⓑ lowest recorded temperature

60. College enrollments Across the United States, community college enrollment grew by 1,400,000 students from 2007 to 2010. In California, community college enrollment declined by 110,171 students from 2009 to 2010. Use integers to write the change in enrollment:

- Ⓐ growth Ⓑ decline

Writing Exercises

61. Give an example of a negative number from your life experience.
62. What are the three uses of the “ $-$ ” sign in algebra? Explain how they differ.

Self Check

Ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
locate positive and negative numbers on the number line.			
order positive and negative numbers.			
find opposites.			
simplify expressions with absolute value.			
translate word phrases to expressions with integers.			

ⓑ If most of your checks were:

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Who can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.